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ABSTRACT

Cross-stock momentum builds on the asymmetry in lead-lag linkages and the difference between long-run and short-run contemporaneous co-movements. Data-driven cross-stock linkages generate a monthly alpha of 1.62% (t -stat=10.03). The asymmetry distinguishes cross-stock momentum from factor momentum, and industry momentum is not subsumed by factor momentum. Factor momentum profit is mostly due to the high cross-stock links. The data-driven linkages vary faster over time than those in previous studies because short-run co-movements incorporate persistent linkages.

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1. Introduction

A growing literature shows that the past return of one stock can predict the stock returns of other connected firms. Such cross-stock return momentum is found among stocks connected by the same industry (Moskowitz and Grinblatt, 1999), the supply chain (Cohen and Frazzini, 2008; Menzly and Ozbas, 2010), and the same geographic

location (Parsons et al., 2020; Jin and Li, 2021). The lead-lag relation also appears among firms that use similar technology (Lee et al., 2019), share the same analyst (Ali and Hirshleifer, 2020), produce complementary products (Lee et al., 2022), or single- and multi-segment firms in the same industries (Cohen and Lou, 2012). Lo and MacKinlay (1990) and Hou (2007) study whether the returns of large stocks lead those of smaller stocks.

The literature on cross-stock momentum raises the following question: Have previous studies of cross-stock momentum pinpointed the main features of the lead-lag linkages across stocks, or are there important features still hidden in the data? Ehsani and Linnainmaa (2022) discuss the difficulty of distinguishing between factor and pure firm-specific momentum when analyzing factor momentum and own-stock momentum. Arnott et al. (2023) argue that industry momentum stems from factor momentum. Is there any feature of cross-stock momentum that separates it from factor momentum?

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We address these questions in this paper. First, we dissect the cross-stock linkages theoretically. The cross-stock linkages consist of an asymmetric component and a symmetric component. The asymmetric linkages between two stocks consist of two directional lead-lag linkages: a high (low) linkage from the leader (laggard) stock's return to the future return of the laggard (leader) stock. The asymmetric linkages reflect cross-sectional variations in lead-lag linkages. The symmetric linkages depend on the differences between long-run and short-run contemporaneous co-movements, which reflect their time series properties.

The asymmetry in cross-stock linkages separates cross-stock momentum from factor momentum because the cross-stock momentum, if purely from factor momentum, is likely symmetric. The symmetry is due to $Cov(R_t^i, R_{t-1}^j) = Cov(R_t^j, R_{t-1}^i) = \beta_i \beta_j Cov(R_t^f, R_{t-1}^f)$ as long as β does not vary substantially between adjacent months.¹ We confirm that time-varying betas at the monthly frequency do not contribute to cross-stock predictability. Therefore, factor momentum does not imply the cross-stock momentum from the asymmetry, which is about whether individual stock i leads stock j more than j leads i . The difficulty with distinguishing between own-stock momentum and factor momentum is due to the lack of asymmetry in own-stock momentum. Ehsani and Linnainmaa (2022) discuss the difficulty of omitted factors—the momentum exhibited by residual returns can stem from the momentum of omitted factors. The asymmetry can distinguish cross-stock momentum from factor momentum without knowing all relevant factors.

Our empirical analysis uses the principal portfolio (PP) method in Kelly et al., 2023 to extract data-driven cross-stock linkages. We build a prediction matrix using the signals from all stocks in the cross-section to predict future stock returns. The PP method characterizes an optimal portfolio in closed form based on the prediction matrix.

We find significant cross-stock return predictability. The monthly alpha is 1.62% (t -stat=10.03) in the 1926–2018 sample period. The cross-stock momentums in previous studies do not subsume the predictability. Therefore, the data-driven approach uncovers important cross-stock linkages new to the literature.

We explore features of the data-driven linkages and relate them to our theoretical characterization to understand their economic implications. First, we use the asymmetry feature to distinguish cross-stock momentum from factor momentum. The asymmetry generates a monthly alpha of 1.02% (t -stat=4.75), which is statistically and economically significant compared to the monthly alpha of 1.95% (t -stat=8.68) from the full prediction matrix in the same sample period when factor momentum can be computed. The evidence indicates that cross-stock momentum is not entirely driven by factor momentum, including momentum of potentially omitted factors.

Arnott et al. (2023) argue that factor momentum subsumes industry momentum. However, we find significant alphas using the asymmetry of industry linkages. How

do we reconcile these findings? We show that the value-weighted industry returns used by Arnott et al., 2023 amplify the misspecification in the asymmetry of industry linkages. Industry linkages emphasize size for the asymmetry, where larger stocks matter more for predicting the future returns of other stocks in the same industry than smaller stocks. However, the predictability may not be proportional to size. The misspecification for large stocks may overwhelm the valid information in asymmetry when amplified by value weight. When we mitigate the misspecification by weighting the largest stocks less, industry momentum becomes significant when replicating the analyses in Arnott et al. (2023).

We also study how cross-stock momentum contributes to factor momentum. We use the same analysis as Ehsani and Linnainmaa (2022) and Arnott et al. (2023) to study the source of factor momentum profits, except that we disaggregate further to the stock level. Lo and MacKinlay (1990) use similar analyses to analyze portfolio lead-lag relations. Exploiting the asymmetry in cross-stock linkages, we find that the factor momentum profit is mostly derived from the high cross-stock links. For example, for time-series factor momentum using the factors in Ehsani and Linnainmaa (2022), the factor momentum profit is 1.10 (t -stat=3.78), -0.68 (t -stat= -2.53), and 0.01 (t -stat=0.63), respectively, from the high cross-stock links, low cross-stock links, and own-stock momentum. Own-stock momentum has a negligible contribution to factor momentum, consistent with Ehsani and Linnainmaa (2022). Under the null hypothesis that cross-stock linkages do not matter for factor momentum or those cross-stock linkages affecting factor momentum are themselves derived from factor momentum, we expect no difference between high and low cross-stock links because factor momentum likely implies symmetric linkages across stocks. Our evidence shows that cross-stock linkages contribute substantially to factor momentum.

Second, we show a time-varying feature of cross-stock linkages. The data-driven linkages are substantially less persistent than the cross-stock linkages identified in previous studies. The data-driven linkage has a five-year autocorrelation of only 2%. By contrast, among the other linkages, common geographic location is the most stable, with a five-year autocorrelation of 95%. This is followed by common industry, with a five-year autocorrelation of 88%. Even the fastest time-varying linkage in previous studies, technological similarity, has a five-year autocorrelation of 43%. The time-varying linkage contributes substantially to the cross-stock momentum. Despite the high persistence of the cross-stock linkages in previous studies, the bulk of their return predictability comes from stock pairs with high data-driven linkages, and such stock pairs have low persistence over time.

The time-varying feature can be explained by the cancellation of persistent components between long-run and short-run co-movements in our theoretical characterization. For example, for industry linkages, both the long-run and short-run co-movements have five-year autocorrelation coefficients of around 0.6. However, the difference between long-run and short-run co-movements has a near-zero five-year autocorrelation. This implies that the per-

¹ R and β denote the return and factor loading for stocks i and j . f is a factor.

sistent components in long-run co-movements are already reflected in short-run co-movements, which suggests that investors recognize those linkages that do not vary much over time. Therefore, predictability is likely from time-varying linkages, consistent with our empirical findings.

We also conduct a dimension reduction of the data-driven linkage using singular value decomposition (SVD). The SVDs are ordered such that the first few terms capture the most important features. Empirically, the leading terms in SVD have higher correlations with asymmetry and the difference between long- and short-run co-movements than the other SVD terms, which validates our theoretical characterization. Consistently, the PP returns from the first few SVDs are sizable compared to the full PP return.

Our paper makes the following contributions to the literature. First, the asymmetric component in our theoretical characterization helps distinguish cross-stock momentum from factor momentum, even with omitted factors. Our results based on the asymmetry show that cross-stock momentum is not entirely driven by factor momentum. [Arnott et al., 2023](#) find that factor momentum subsumes industry momentum. However, our evidence suggests that their finding is likely due to the misspecification in the asymmetry of industry linkages, particularly those associated with large stocks, amplified by value-weighted industry returns. After mitigating the misspecification, industry momentum becomes significant when we replicate their analysis. Our results suggest a more prominent role for asymmetry when specifying cross-stock linkages, which can potentially enhance cross-stock momentum and help differentiate with factor momentum.

Second, we find that the profits of factor momentum in [Ehsani and Linnainmaa \(2022\)](#) and [Arnott et al., 2023](#) are mostly from high cross-stock links. [Lo and MacKinlay \(1990, p. 184\)](#) point out the possibility that positive autocorrelation in index returns “may be attributed solely to the positive cross-autocorrelations across securities” (see their Eq. (10) and discussion afterward). We contribute relative to [Lo and MacKinlay \(1990\)](#) by exploiting the asymmetry in cross-stock linkages. Asymmetry provides evidence that the cross-autocorrelations responsible for factor momentum are not due to factor momentum because the cross-autocorrelations derived from factor momentum are likely symmetric. Our finding contributes to the understanding of factor momentum.

Third, we find evidence that the persistent parts of long-run co-movements are incorporated into the short-run co-movements, which is consistent with investors recognizing the persistent co-movements. Our data-driven linkages vary over time at approximately the business cycle frequency. In contrast, the cross-stock linkages in previous studies—including linkages based on shared analysts ([Ali and Hirshleifer, 2020](#)), customer and supplier in the supply chain ([Menzly and Ozbas, 2010](#)), complicated firms ([Cohen and Lou, 2012](#)), economic link ([Cohen and Frazzini, 2008](#)), common geographic location ([Parsons et al., 2020](#)), same industry ([Moskowitz and Grinblatt, 1999](#)), or similar technology ([Lee et al., 2019](#))—are specified from economic insights and highly persistent over time. Our results contribute to the cross-stock momentum literature

by pinpointing the time-varying feature of the cross-stock linkages responsible for return alphas.

Fourth, we contribute to the asset pricing literature using machine learning. We use the PP in [Kelly et al., 2023](#) for feature extraction and dimension reduction. We point out that the usual portfolio construction in the cross-stock momentum literature based on return forecasts corresponds to PP under the Schatten 2-norm, while the optimal PP is under the Schatten 1-norm. Therefore, PP provides a general framework for the empirical analysis of networks between firms. [Kelly et al., 2023](#) also analyze the symmetry and asymmetry of the prediction matrix. Building on their insight, we further decompose the symmetric component into long-run and short-run co-movements to understand how the time series properties of co-movements induce time-varying linkages. We also decompose the asymmetric component into high and low cross-stock links and show that the high cross-stock links are mainly responsible for factor momentum. The dimension reduction validates that the time-series properties of co-movements and the cross-sectional asymmetry are important features, contributing to a parsimonious representation of cross-stock linkages advocated by [Ali and Hirshleifer, 2020](#). Our analysis of the cross-stock prediction matrix potentially has broad applicability for studying firm networks, which are often modeled by adjacency matrices ([Jackson, 2008](#)).

There are other machine learning techniques. Some recent studies implement principal component analysis or its extensions, such as [Kelly et al. \(2019\)](#), [Lettau and Pelger \(2020\)](#), and [Kim et al. \(2020\)](#). [Rapach et al. \(2013\)](#), [Chinco et al. \(2019\)](#), [Freyberger et al. \(2020\)](#), and [Feng et al. \(2020\)](#) use LASSO or its variations. [Gu, Kelly, Xiu, 2020](#) and [Rossi \(2018\)](#) employ neural networks and trees. [Kozak et al. \(2020\)](#) use shrinkage. [Scherbina and Schlusche \(2020\)](#) run a lead-lag return regression for each stock pair. We acknowledge that other methods might also provide adequate feature extraction. Overall, we find that PP helps to characterize the cross-stock linkages and the optimal portfolio simultaneously.

The rest of the paper is organized as follows. [Section 2](#) theoretically characterizes the sources of cross-stock momentum. [Section 3](#) discusses the PP method. [Section 4](#) describes the data. [Section 5](#) presents the cross-stock momentum return from the data-driven linkages. We analyze the relation between cross-stock momentum and factor momentum in [Section 6](#). [Section 7](#) studies the time variation of cross-stock linkages. We conclude in [Section 8](#).

2. Dissecting cross-stock momentum

In this section, we theoretically characterize the lead-lag linkages. First, we define the notations for the lead-lag relation in months $t - 1$ and t . Let r_t^i denote the continuously compounded return of stock i in month t . For stock i , $\sigma_{i,t}$ denotes its return volatility in month t . Without time subscript, σ_i denotes its monthly return volatility during the months $t - 1$ and t . $\sigma_i(2)$ denotes the volatility of two-month returns from $t - 1$ to t for stock i . $VR_i \equiv \frac{\sigma_i^2(2)}{2\sigma_i^2}$ denotes the variance ratio between the two-

and one-month returns. $Cov(x, y)$ and $Cor(x, y)$ denote, respectively, the covariance and the correlation between variables x and y . $COV(2) \equiv Cov(r_t^i + r_{t-1}^i, r_t^j + r_{t-1}^j)$ and $COR(2) \equiv Cor(r_t^i + r_{t-1}^i, r_t^j + r_{t-1}^j)$ denote the covariance and correlation, respectively, of two-month returns between stocks i and j . $COV(1) \equiv (Cov(r_t^i, r_t^j) + Cov(r_{t-1}^i, r_{t-1}^j))/2$ and $COR(1) \equiv (Cor(r_t^i, r_t^j) + Cor(r_{t-1}^i, r_{t-1}^j))/2$ denote the covariance and correlation, respectively, of monthly returns between stocks i and j during the months $t-1$ and t on average. Next, Proposition 1 characterizes the lead-lag linkages across stocks.

Proposition 1. *The lead-lag linkage across two stocks i and j comprises a symmetric component (SYM) and an asymmetric component (ASY), where $SYM_{i,j} = SYM_{j,i}$ and $ASY_{i,j} = -ASY_{j,i}$. The lead-lag linkage satisfies the following:*

$$\begin{aligned} Cov(r_t^i, r_{t-1}^j) &= SYM_{i,j} + ASY_{i,j} \\ Cov(r_t^j, r_{t-1}^i) &= SYM_{j,i} + ASY_{j,i} = SYM_{i,j} - ASY_{i,j}, \end{aligned} \quad (1)$$

where SYM and ASY satisfy:

$$SYM_{i,j} = \frac{1}{2} \underbrace{[COV(2) - 2COV(1)]}_{\text{long-short cov}} \quad (2)$$

$$\begin{aligned} &= COR(2)\sigma_i\sigma_j\sqrt{VR_i VR_j} - \frac{1}{2}Cor(r_t^i, r_t^j)\sigma_{i,t}\sigma_{j,t} \\ &\quad - \frac{1}{2}Cor(r_{t-1}^i, r_{t-1}^j)\sigma_{i,t-1}\sigma_{j,t-1} \end{aligned} \quad (3)$$

$$\approx \underbrace{[COR(2) - COR(1)]}_{\text{long-short cor}} \sigma_i \sigma_j. \quad (4)$$

$$ASY_{i,j} = \frac{1}{2} \underbrace{[Cov(r_t^i, r_{t-1}^j) - Cov(r_{t-1}^i, r_t^j)]}_{\text{asymmetric cov}} \quad (5)$$

$$\approx \frac{1}{2} \underbrace{[Cor(r_t^i, r_{t-1}^j) - Cor(r_{t-1}^i, r_t^j)]}_{\text{asymmetric cor}} \sigma_i \sigma_j. \quad (6)$$

The approximation in Eq. (6) holds if $\sigma_{i,t} \approx \sigma_{i,t-1} \approx \sigma_i$ and $\sigma_{j,t} \approx \sigma_{j,t-1} \approx \sigma_j$. The approximation in Eq. (4) holds under the additional assumption that $VR_i \approx VR_j \approx 1$.

The symmetric component, SYM, depends on the long-run co-movement $COV(2)$ minus the short-run co-movement $2COV(1)$. The difference, $COV(2) - 2COV(1)$, for cross-stock momentum is analogous to the variance ratio for own-stock momentum. Campbell et al. (1997, Chap. 2.4.3) show that the return autocovariance can be inferred by comparing the variance of two-month returns with twice the variance of one-month returns because a higher autocovariance implies a higher variance ratio. Equation (2) shows that a higher cross-stock momentum is associated with a higher $COV(2)$ relative to $2COV(1)$.²

$COV(2)$ and $COV(1)$ are influenced by own-stock momentum (through the variance ratio VR_i and VR_j in Eq. (3)) and time-varying volatility. To disentangle such influences in the empirical analyses, Eqs. (4) and (6) provide approximate formulae using correlations. Our findings are robust whether using correlation or covariance. Equation (4) compares $COR(2)$ to $COR(1)$ instead of $2COR(1)$ because the effect of return horizon on variance—two-month return variance should be compared to twice the one-month return variance—is eliminated in correlation.

If $COR(2) \neq 0$, the two stocks receive correlated shocks. But if the returns of the two stocks fully reflect the correlated shocks in the same month, i.e., $COR(1) = COR(2)$, there is no lead-lag relation (approximately) for the symmetric component, as shown in Eq. (4).

The asymmetric component, ASY, reflects whether stock j leads stock i more than stock i leads stock j . Given two stocks i and j , we define the leader stock of the pair as follows.

Definition 1. Stock j leads stock i (equivalently, i lags j) if $Cor(r_t^i, r_{t-1}^j) > Cor(r_t^j, r_{t-1}^i)$.

To simplify terminology, “ j leads i ” indicates that the directional linkage from r_{t-1}^j to r_t^i is more positive than the directional linkage from r_{t-1}^i to r_t^j .³ In Definition 1, we compare correlation. Alternatively, leader stocks can be defined by comparing covariance, which is affected by time-varying volatility. Our results are robust to the alternative definition using covariance.

For own-stock momentum, Proposition 1 reduces to Corollary 1.

Corollary 1. *If the two stocks are the same, i.e., $i = j$, SYM and ASY become:*

$$\begin{aligned} SYM_{i,j} &= SYM_{j,i} = Cov(r_t^i, r_{t-1}^i) \\ ASY_{i,j} &= ASY_{j,i} = 0. \end{aligned}$$

In Corollary 1, we emphasize that cross-stock momentum involves a new concept of asymmetry because ASY does not play a role in own-stock momentum. Ehsani and Linnainmaa (2022) point out the difficulty distinguishing between own-stock momentum and factor momentum. The absence of ASY implies that it cannot contribute to such differentiation. By contrast, in Section 6 we distinguish cross-stock momentum from factor momentum using asymmetry.

Ali and Hirshleifer, 2020 find that adding the short-term reversal factor increases the cross-stock momentum alpha and explain that last month's connected firms' returns correlate with a stock's own return last month. Corollary 2 formalizes a theoretical basis for their empirical findings.

Corollary 2. *If $COR(2) > 0$, the symmetric component of cross-stock momentum (SYM) is increasing in the variance ratios VR_i and VR_j , ceteris paribus.*

² Unlike the variance ratio, we use the difference between $COV(2)$ and $2COV(1)$ instead of their ratio because $COV(1)$ may be zero.

³ Positive $Cor(r_t^i, r_{t-1}^j)$ by itself does not necessarily imply that j leads i as defined in Definition 1.

A stronger short-term reversal lowers SYM by lowering variance ratios. Therefore, controlling for short-term reversal will likely increase the cross-stock momentum alpha in empirical analyses.

3. The PP of cross-stock momentum

Following Kelly et al., 2023, we first construct the prediction matrix $\hat{\Pi}$, which relates the stock returns to the lagged signals across all stocks using data from rolling T -month windows:

$$\hat{\Pi}_t = \frac{1}{T} \sum_{\tau=t-T+1}^t R_\tau S'_{\tau-1}, \quad (7)$$

where R_τ is a vector containing the individual stock simple returns in month τ , which are cross-sectionally demeaned to focus on cross-sectional return predictability. $S_{\tau-1}$ is a vector of signals for predicting R_τ . We use the one-month returns across all stocks in month $\tau - 1$ as $S_{\tau-1}$. The signals contained in $S_{\tau-1}$ are rank transformed to within the range $[-0.5, 0.5]$ to avoid excessive influence by outliers.⁴ This is a rolling implementation of the PP method in Kelly et al., 2023.

Kelly et al., 2023 consider a linear strategy based on the signal S , which has portfolio weights of the following form:

$$w_t = L' S_t. \quad (8)$$

The matrix L' translates the signal into the portfolio weight vector w .⁵ The optimal linear strategy builds on the singular value decomposition (SVD) of the prediction matrix $\hat{\Pi}$:

$$\hat{\Pi} = U \cdot \Lambda \cdot V' = \sum_{k=1}^n \lambda_k u_k v'_k. \quad (9)$$

SVD decomposes the prediction matrix into a product of three matrices: U , Λ , and V' . u and v are the singular vectors. Different u vectors are orthogonal to each other. Similarly, different v vectors are orthogonal to each other. u_k and v_k each have a vector norm of one. The columns of matrix U (or V) contain the singular vectors u (or v). Λ is a diagonal matrix whose diagonal contains the singular values λ_k .

An interesting implication of SVD is that the lead-lag relation across firms has $O(n)$ complexity instead of $O(n^2)$. The n terms in Eq. (9) suffice to represent $\hat{\Pi}$, due to Λ being diagonal. The singular value λ_k represents the importance of each term. The reduced complexity of $O(n)$ implies that using parsimonious models to explain cross-stock linkages, suggested by Ali and Hirshleifer, 2020, is feasible ex ante. This is an exact dimension reduction.

SVD also allows an approximate dimension reduction using a few of the largest singular values. The singular values $\lambda_1, \lambda_2, \dots, \lambda_n$ are in descending order, such that

λ_1 is the largest singular value. The approximate prediction matrix using the largest $K \leq n$ singular values is: $\hat{\Pi} \approx \sum_{k=1}^K \lambda_k u_k v'_k$.

Kelly et al., 2023 show that the optimal linear strategy builds on the singular vectors. Specifically, the optimal L in Eq. (8) satisfies:

$$L = \sum_k v_k u'_k. \quad (10)$$

Therefore, the optimal PP return using the K largest singular values is:

$$w'_t R_{t+1} = S'_t L R_{t+1} = S'_t \sum_{k=1}^K (v_{k,t-1} u'_{k,t-1}) R_{t+1}. \quad (11)$$

The prediction matrix constructed from the linear combination of T -month data has at most T non-zero singular values. Therefore, we use $K \leq T$ in the empirical analysis to exclude zero singular values. Terms associated with zero singular values do not improve the portfolio expected return (Kelly et al., 2023, Proposition 11). This is also consistent with the shrinkage in Kelly et al., 2023 using only large singular values. Following Kelly et al., 2023, the portfolio weights are further scaled by a multiplicative constant so that the portfolio return volatility matches the market volatility.

In Eq. (11), we lag the prediction matrix by one month and use singular vectors of $\hat{\Pi}_{t-1}$, instead of $\hat{\Pi}_t$, to construct the portfolio weights w_t . Since the signal S_t is one-month stock returns, S_t does not overlap the returns we use to construct $\hat{\Pi}_{t-1}$. Avoiding the overlap improves the prediction matrix estimate, as discussed in Section A.8 of the Online Appendix. When the signal S_t uses one-month returns, the overlap is small, and we obtain similar results from $\hat{\Pi}_t$ and $\hat{\Pi}_{t-1}$. Nonetheless, we lag the prediction matrix by one month in Eq. (11) to be consistent.

The PP provides a general framework for analyzing portfolios based on cross-stock linkages. In particular, it nests the portfolio sorts typically used in cross-stock momentum studies as a special case. In portfolio sorts, the stocks are sorted based on their predicted future returns to create long-short portfolios. The return predictions are obtained by multiplying the prediction matrix by the signal vector.⁶ Therefore, the portfolio weights are determined by the predicted return $\hat{\Pi} S_t$. Comparing $\hat{\Pi} S_t$ with Eq. (8) shows that a portfolio based on the predicted returns corresponds to a linear strategy with

$$L = \hat{\Pi}' = \sum_k \lambda_k v_k u'_k, \quad (12)$$

where the last step of Eq. (12) follows from the SVD of $\hat{\Pi}$ in Eq. (9). The linear strategy in Eq. (12) resembles the optimal L in Eq. (10), other than weighting by the singular values. Proposition 2 characterizes the relation between the optimal PP and the portfolio based on predicted returns.

Proposition 2. The portfolio return $S'_t L R_{t+1}$, where $L = \hat{\Pi}' = \sum_k \lambda_k v_k u'_k$, is equivalent to the optimal PP return under

⁴ Rank transformation is common in machine learning and has been used by Kelly et al. (2019), Freyberger et al. (2020), Gu, Kelly, Xiu, 2020, and Kelly et al., 2021, among others. Our results are similar if we use raw signals, although the results from rank-transformed signals are stronger.

⁵ To simplify the notation, we do not make the dependence of L on time explicit.

⁶ To simplify the terminology, we also refer to the matrices characterizing the cross-stock linkages in other studies as prediction matrices.

the Schatten 2-norm corresponding to $q = 2$ in the general solution in Kelly et al., 2023. The optimal PP return in Kelly et al., 2023 with $L = \sum_k v_k u'_k$ is under the Schatten 1-norm corresponding to $q = 1$. The expected return of the optimal PP is higher under the 1-norm than the 2-norm.

Proposition 2 shows that the optimal PP outperforms the Schatten 2-norm PP, which is equivalent to using the predicted returns as portfolio weights. A long-short portfolio based on the predicted returns typically includes an extra step where the stocks are sorted into quantiles, such as deciles, based on the predicted returns. In numerical analyses, we find that the extra step of decile sort does not affect the returns much. Therefore, a long-short portfolio sorted by the predicted returns is approximately the Schatten 2-norm PP.

4. Data and sample

Stock returns and other relevant market data from the 1926 to 2018 sample period come from the CRSP monthly files. We restrict our analysis to common stocks (share codes 10 and 11) from the NYSE, AMEX, and NASDAQ. Following Fama and French (2008) and Hou et al., 2020, we exclude microcap stocks whose market capitalization is below the 20th percentile of the NYSE. To avoid delisting bias, we use the correction approach in Shumway (1997). We obtain the time series of Standard Industrial Classification (SIC) codes from CRSP. If the SIC code is missing in CRSP, we use the most recent SIC code from Compustat. Accounting data, geographic location, and industry segment are from Compustat. We follow the standard convention and assume that the accounting data for fiscal years ending in year t become available at the end of June in year $t + 1$. We obtain the factor returns from Kenneth French's website.⁷

We obtain the annual input-output data from the Bureau of Economic Analysis (BEA). Following Menzly and Ozbas (2010), we draw on 12 different surveys (2007, 2002, 1997, 1992, 1987, 1982, 1977, 1972, 1967, 1963, 1958, 1947) to measure the flow of goods to and from industries. Specifically, we use the survey data starting from the end of the years in which the surveys are publicly released (2012, 2007, 2002, 1997, 1994, 1991, 1984, 1979, 1974, 1969, 1964, 1952). The pre-1997 BEA surveys use the SIC industry classification. The surveys in 1997 and later use the North American Industry Classification System (NAICS). We use the historical NAICS codes from Compustat if available; if not available, we use the NAICS codes from CRSP. For the customer-supplier links, we follow Cohen and Frazzini (2008) and lag the customer-supplier link data by six months.

We use the Google patent data provided by Kogan et al. (2017) to construct technology linkages. Following Lee et al. (2019), we exclude financial stocks from the analysis and assume that the patent data become publicly available six months after the end of the patent

grant year. The sample period for the technology link momentum ends in 2012 due to patent data availability.

We use the Institutional Brokers Estimate System (IBES) detail file to identify analyst co-coverage. Following Ali and Hirshleifer, 2020, we define two stocks as "connected" if at least one analyst covers both stocks. We determine that an analyst covers a stock if the analyst issues at least one FY1 or FY2 earnings forecast over the past 12 months.

Table 1 presents the summary statistics. On average, the sample covers 86% of the total market capitalization in the CRSP universe. There are 3265 firms in a typical cross-section in CRSP. After excluding microcap stocks, 1417 firms remain on average. We use rolling five-year windows to construct the prediction matrix. In Section A.2 of the Online Appendix, we examine other rolling windows. When we use five-year rolling windows to construct the prediction matrix, requiring non-missing monthly returns in the rolling window leaves 848 firms on average in the final cross-section. We can compute the prediction matrix for the 1931–2018 sample period.

5. Cross-stock momentum using data-driven linkages

Table 2 shows the cross-stock momentum returns using PP. The optimal PP has a monthly return of 1.16% (t -stat = 7.15) in excess of the risk-free rate. The optimal PP also has positive alphas relative to the CAPM, and three- and four-factor models. The three-factor model, denoted FF3, includes the market, size, and value factors. The four-factor model additionally includes the momentum factor, and is denoted FF3+m. The monthly four-factor alpha is 1.17% (t -stat = 6.96).

The alpha is even bigger from a five-factor model that additionally controls for the short-term reversal factor, which is denoted FF3+mr. The monthly five-factor alpha is 1.62% (t -stat = 10.03). Adding the short-term reversal factor increases the cross-stock momentum alpha, consistent with Corollary 2.

Table 2 also shows the alpha relative to the Fama and French (2015) five-factor model, denoted FF5. Due to the data availability of the FF5 factors, the sample period starts in 1963. The monthly FF5 alpha is 1.35% (t -stat = 5.49). When we additionally control for the momentum and short-term reversal factors in a seven-factor model (denoted FF7), the monthly alpha is even higher at 1.95% (t -stat = 8.68).⁸ The cross-stock predictability is not driven by the cross-stock momentums found in previous studies, as shown in Section A.3 of the Online Appendix.

The optimal PP in Table 2 uses all non-zero singular values of the prediction matrix. Table 2 also shows the returns using the first few PPs with the largest singular values. The first PP alone has a seven-factor alpha of 1.08% (t -stat = 4.79). The alpha rises to 1.37% (t -stat = 6.02) when using the first five PPs. The first twenty PPs generate a seven-factor alpha of 1.81% (t -stat = 8.07), capturing 93% of the optimal PP alpha. Table 2 also shows that the optimal PP, under the Schatten 1-norm, outperforms the 2-norm

⁷ The URL is https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁸ Table A1 of the Online Appendix shows the alphas using the same post-1963 sample period for all the factor models. All the alphas are significant.

Table 1
Summary statistics.

This table reports summary statistics on the number of firms (# firms), the average market capitalization (size, in millions), the total market capitalization as a fraction of the total CRSP value, and the monthly individual-stock returns used by PP and other cross-stock momentums. Panel B additionally shows the remaining number of firms and firm size after applying filters to the full CRSP universe of common stocks on the NYSE, AMEX, and NASDAQ. The filters include: (1) excluding microcap stocks whose sizes are below the 20th NYSE size percentile; and (2) non-missing monthly return data in the past five years.

Panel A.							
	Mean	Std. Dev.	Min	25%	Median	75%	Max
Number of Firms	848	314	237	637	854	1158	1289
Fraction of CRSP value (%)	86.23	4.36	71.45	83.77	86.62	88.89	95.07
Ret (%)	1.19	8.06	−31.44	−3.56	0.76	5.41	50.35
Panel B.							
	Avg. # firms		Avg. size		Sample period		
PP	848		3680		1931–2018		
CRSP universe	3265		1138				
after: exclude microcap	1417		2385				
after: require 5-year ret	848		3680				
Analyst Coverage	3658		3266		1984–2018		
BEACustomer	3135		1445		1953–2018		
BEASupplier	3135		1445		1953–2018		
Complicated Firms	848		5140		1977–2018		
Economic Link	2747		2422		1977–2018		
Geographic Link	2687		2024		1970–2018		
Industry Momentum	3265		1138		1931–2018		
Technology Link	927		2802		1963–2012		

Table 2
Return of cross-stock momentum.

This table shows the PP returns based on the cross-stock momentum. Following Kelly et al. (2023), we scale the PP portfolio weights by a multiplicative constant to match the market volatility. The PP prediction matrix uses rolling 60-month windows. The optimal PP return uses the first 60 PP terms. This table also shows the returns from using only the first 1, 3, 5, 10, or 20 PP terms, as well as the Schatten 2-norm PP returns in Proposition 2. This table reports the monthly excess return over risk-free rate and the alphas relative to the CAPM, the Fama and French (1993) three-factor model (FF3), the FF3 model augmented with the Carhart (1997) momentum factor (FF3+m), the FF3 model augmented with the momentum and short-term reversal factors (FF3+mr), the Fama and French (2015) five-factor model (FF5), and the FF5 model augmented with the momentum and short-term reversal factors (FF7). *t*-statistics are in the parentheses. The sample period is 1931–2018 except when using the FF5 factors (1963–2018).

	Excess returns	CAPM alpha	FF3 alpha	FF3 + m alpha	FF3 + mr alpha	FF5 alpha	FF7 alpha
Optimal PP	1.16 (7.15)	1.17 (7.13)	1.15 (6.97)	1.17 (6.96)	1.62 (10.03)	1.35 (5.49)	1.95 (8.68)
PP1-1	0.51 (3.11)	0.51 (3.14)	0.48 (2.92)	0.53 (3.16)	0.78 (4.64)	0.69 (2.98)	1.08 (4.79)
PP1-3	0.65 (3.99)	0.65 (3.96)	0.61 (3.74)	0.62 (3.69)	0.92 (5.51)	0.77 (3.23)	1.18 (5.15)
PP1-5	0.76 (4.70)	0.76 (4.65)	0.73 (4.44)	0.77 (4.61)	1.10 (6.63)	0.83 (3.41)	1.37 (6.02)
PP1-10	0.93 (5.71)	0.90 (5.52)	0.86 (5.28)	0.92 (5.47)	1.25 (7.55)	1.00 (4.13)	1.59 (7.03)
PP1-20	1.05 (6.47)	1.04 (6.32)	1.00 (6.09)	1.06 (6.32)	1.45 (8.82)	1.18 (4.81)	1.81 (8.07)
2-norm PP	0.80 (4.93)	0.81 (4.91)	0.78 (4.74)	0.88 (5.22)	1.27 (7.74)	1.08 (4.19)	1.71 (7.18)

PP. This is consistent with Proposition 2 and confirms that the optimal PP is better than the portfolio whose weights are determined solely by the expected return forecasts of individual stocks.

Fig. 1 shows the cumulative gross return from a \$1 initial investment in the optimal PP, which performs well throughout the sample period. The performance indicates that the prediction matrix captures valuable information on cross-stock linkages.

Table 3 shows the PP returns from the asymmetric and symmetric components of the cross-stock linkage in Proposition 1. When using the asymmetric component ASY, the FF7 alpha is 1.02% (*t*-stat = 4.75). The symmetric component SYM delivers an FF7 alpha of 1.33% (*t*-stat = 6.02). Therefore, both components contribute to cross-stock momentum.

Table 3 also shows that the asymmetric correlation and the difference in long-term and short-term contemporane-

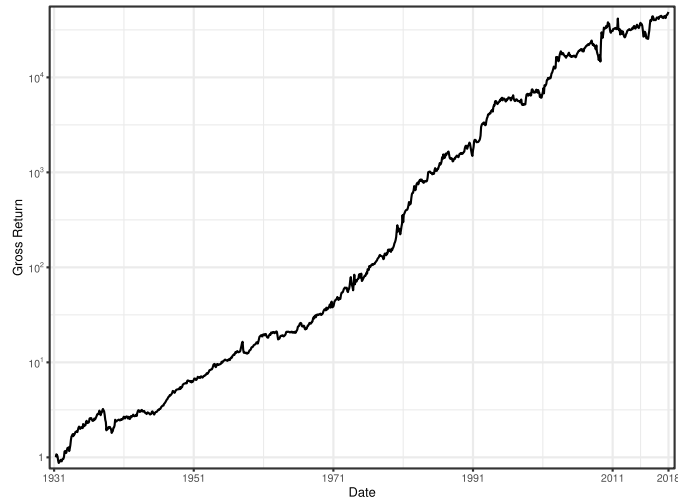


Fig. 1. Cumulative return of PP based on cross-stock momentum. This figure shows the gross cumulative returns from a \$1 initial investment in the PP in 1931–2018.

Table 3

Asymmetric and symmetric components of cross-stock linkages.

This table is similar to Table 2 except that we use the asymmetric or symmetric components in Proposition 1 as the prediction matrix. The asymmetric components include “asymmetric cor” and “asymmetric cor” multiplied by $\sigma_i\sigma_j$ in Eq. (6), and the full asymmetric component ASY in Eq. (5). The symmetric components include “long-short cor” and “long-short cor” multiplied by $\sigma_i\sigma_j$ in Eq. (4), and the full symmetric component SYM in Eq. (3).

	Excess returns	CAPM alpha	FF3 alpha	FF3 + m alpha	FF3 + mr alpha	FF5 alpha	FF7 alpha
Asymmetric							
asymmetric cor	0.56 (3.47)	0.46 (2.86)	0.43 (2.65)	0.95 (6.36)	0.80 (5.26)	0.57 (2.30)	1.07 (5.02)
asymmetric cor · $\sigma_i\sigma_j$	0.50 (3.08)	0.41 (2.51)	0.38 (2.35)	0.88 (5.82)	0.72 (4.67)	0.55 (2.22)	1.04 (4.87)
ASY	0.49 (3.04)	0.40 (2.46)	0.37 (2.30)	0.87 (5.72)	0.71 (4.58)	0.53 (2.17)	1.02 (4.75)
Symmetric							
long-short cor	0.75 (4.69)	0.73 (4.54)	0.69 (4.27)	0.89 (5.46)	1.20 (7.46)	0.93 (4.00)	1.50 (6.87)
long-short cor · $\sigma_i\sigma_j$	0.63 (3.95)	0.61 (3.81)	0.58 (3.62)	0.76 (4.66)	1.05 (6.42)	0.82 (3.49)	1.37 (6.17)
SYM	0.61 (3.86)	0.60 (3.77)	0.58 (3.61)	0.74 (4.48)	1.00 (6.09)	0.80 (3.43)	1.33 (6.02)

ous correlation essentially capture all the cross-stock predictability in ASY and SYM. Using the asymmetric correlation in Eq. (6) as the prediction matrix generates an FF7 alpha of 1.07% (t -stat = 5.02), comparable to the FF7 alphas from using ASY or the product of asymmetric correlation and $\sigma_i\sigma_j$. Similarly, using the long-short cor in Eq. (4) results in an FF7 alpha comparable to those from using SYM or the product of long-short cor and $\sigma_i\sigma_j$.

Fig. 2 shows the correlation between each SVD term of the prediction matrix and the components of cross-stock momentum characterized in Proposition 1. The leading SVD terms have substantially higher correlations with the cross-sectional asymmetry (asymmetric cor in Eq. (6)) and the time series properties of co-movements (long-short cor in Eq. (4)) than the later SVD terms. Therefore, machine learning confirms the key ingredients of cross-stock momentum prescribed by theory. By contrast, the leading SVD

terms have low correlations with volatility and the variance ratio, which primarily reflect contemporaneous price fluctuations and own-stock momentum instead of cross-stock momentum.

6. Factor momentum

Let F_t denote the return of a factor at time t . Ehsani and Linnainmaa (2022) and Arnott et al., 2023 show that F_{t+1} tends to be higher if the lagged factor return is higher.⁹ We first examine whether cross-stock momentum is robust to controlling for factor momentum. We then analyze whether cross-stock momentum contributes to factor momentum.

⁹ Lewellen (2002) documents momentum for the size and B/M portfolios.

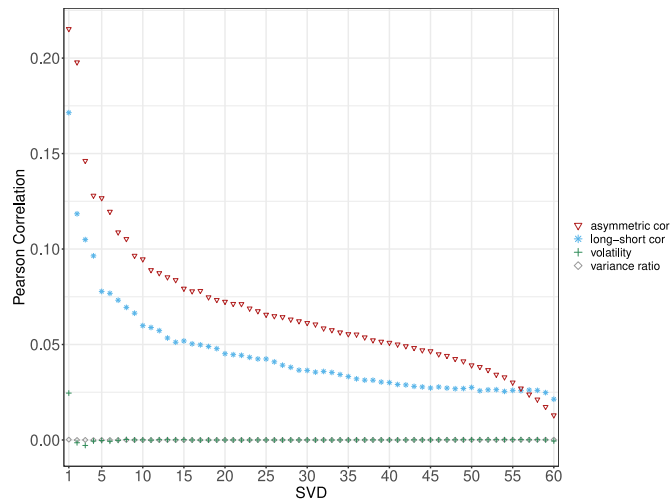


Fig. 2. SVD and the components of cross-stock linkages. This figure shows the correlations between the SVD of the prediction matrix and the components of cross-stock linkages characterized in Proposition 1. For each SVD term $u_k v_k'$, we calculate its correlation with “asymmetric cor” in Eq. (6), “long-short cor” in Eq. (4), and $\sigma_{i,t} \sigma_{j,t}$ and $\sqrt{VR_i VR_j}$ in Eq. (3) (labeled “volatility” and “variance ratio” in the plot, respectively). We compute the correlation for each month and then average it over time. The matrices are vectorized for computing the correlations. This figure shows the correlation for the first 60 SVD components. The sample period is 1931–2018.

Table 4

PP return of cross-stock momentum, controlling for factor momentum.

Panel A is similar to Table 2 except that we control for factor momentum in addition to the FF5 and FF7 controls. We use the 15 factors in Ehsani and Linnainmaa (2022) or the 43 factors in Arnott et al. (2023) to construct factor momentum. We use three different cutoffs to define top factors (long) and bottom factors (short). “TopBottom” selects the top 15% and bottom 15% factors (rounded to the nearest integer) based on past factor returns, following Moskowitz and Grinblatt (1999). Following Ehsani and Linnainmaa (2022), “CrossSection” defines top/bottom factors by comparing past factor returns to the cross-sectional median, while “TimeSeries” defines top/bottom factors as positive/negative past factor returns. Panel B is similar to Table 2 except that we use the residual returns relative to all 43 factors in Arnott et al. (2023) to construct the prediction matrix in Eq. (7). We also use the residual returns for the signals and holding returns (S_t and R_{t+1} in Eq. (11)). The residuals are calculated monthly from a cross-sectional regression of stock returns on all 43 characteristics. Due to the availability of factor data, the sample period is 1963–2018.

Panel A. Control factor momentum				
Controlled Factor Mom		FF5 alpha	FF7 alpha	
15 factors	TopBottom	0.85 (3.95)	1.29 (5.79)	
	CrossSection	0.77 (3.54)	1.25 (5.5)	
	TimeSeries	0.92 (4.14)	1.46 (6.35)	
43 factors	TopBottom	0.83 (3.86)	1.25 (5.49)	
	CrossSection	0.81 (3.69)	1.34 (5.69)	
	TimeSeries	0.83 (3.8)	1.33 (5.73)	
Panel B.Cross-stock momentum using residual returns for prediction matrix, signals, and holding returns				
Excess	CAPM	FF3	FF5	FF7
returns	alpha	alpha	alpha	alpha
0.80	0.76	0.80	0.87	0.96
(4.37)	(4.12)	(4.30)	(4.53)	(4.91)

6.1. Control for factor momentum

In Panel A of Table 4, we control for the factor momentum return when calculating the FF5 and FF7 alphas in Table 2. We include three methods to construct factor momentum, which differ in the cutoff for defining

high or low past factor returns. The “TopBottom” method selects the top and bottom factors based on past factor returns, following Moskowitz and Grinblatt (1999).¹⁰

¹⁰ Moskowitz and Grinblatt (1999) use the top and bottom three industries among their 20 industry portfolios. When controlling for the 15 fac-

Following Ehsani and Linnainmaa (2022), the “CrossSection” method defines top or bottom factors by comparing past factor returns to the cross-sectional median, while the “TimeSeries” method defines top or bottom factors as those with positive or negative past returns. We use the past one-month factor return to sort top and bottom factors and the returns in the subsequent month to measure the factor momentum. Falck et al., 2022 find that the factor momentum is particularly strong at the one-month lead-lag horizon. They show that factor momentum is spanned by stock momentum and factor exposure except at the one-month time scale, though their analysis does not include cross-stock momentum. The one-month lead-lag horizon, left unexplained by Falck et al., 2022, is the focus of cross-stock momentum studies.

The results in Panel A of Table 4 confirm significant cross-stock momentum after controlling for factor momentum. The results from the three versions of factor momentum are similar. The lowest FF7 alpha after controlling for all 43 factors in Arnott et al., 2023 remains significant at 1.25% per month (t -stat=5.49).

Next, we control for factor momentum differently using only residual returns to construct cross-stock momentum following Burt and Hrdlicka (2021) and Lee et al. (2019). In other words, we use residual returns to construct the prediction matrix in Eq. (7) and use residual returns for the signals and holding returns (S_t and R_{t+1} in Eq. (11)). The residual returns are from monthly cross-sectional regressions of stock returns on loadings of all 43 factors in Arnott et al., 2023. Panel B of Table 4 shows the results. The FF7 alpha remains significant at 0.96% per month (t -stat=4.91).

6.2. Cross-stock momentum from the asymmetric component

In this section, we exploit the asymmetric component described in Section 2 to distinguish cross-stock momentum from factor momentum. The cross-stock momentum, if derived from factor momentum, is likely symmetric. To see this, let R^f denote the return of factor f . Let R^i and R^j denote the returns of two stocks, i and j . The lead-lag relation between the two stocks implied by factor momentum satisfies:

$$\begin{aligned} \text{Cov}(R_t^i, R_{t-1}^j) &= \text{Cov}(\beta_i R_t^f, \beta_j R_{t-1}^f) = \beta_i \beta_j \text{Cov}(R_t^f, R_{t-1}^f) \\ &= \text{Cov}(R_t^j, R_{t-1}^i), \end{aligned} \quad (13)$$

assuming the factor loadings β_i and β_j do not change between adjacent months.¹¹ The lead-lag relation induced by factor momentum is symmetric in the sense that $\text{Cov}(R_t^i, R_{t-1}^j) = \text{Cov}(R_t^j, R_{t-1}^i)$. Therefore, the symmetric component *SYM* in Proposition 1 captures the potential effect of factor momentum, including the momentum of omitted factors.

tors in Ehsani and Linnainmaa (2022), the “TopBottom” method in Panel A of Table 4 selects top or bottom $3/20 \times 15 \approx 2$ factors. When controlling for the 43 factors in Arnott et al., 2023, the “TopBottom” method selects top or bottom $3/20 \times 43 \approx 6$ factors.

¹¹ Section A.1 of the Online Appendix shows that the asymmetry induced by time-varying betas and stocks' weights in the factors at the monthly horizon is insufficient to affect our results.

The cross-stock momentum derived from the asymmetric component *ASY* is unrelated to factor momentum. Because $ASY_{i,j} + ASY_{j,i} = 0$ according to Proposition 1, the average lead-lag relation between stocks i and j in *ASY* has no exposure to factor momentum. Instead, comparing $ASY_{i,j}$ and $ASY_{j,i}$ indicates whether stock j leads or lags stock i according to Definition 1 and Eq. (6). Factor momentum does not predict whether one stock leads or lags another stock as in Definition 1. Using only the asymmetric component likely gives a conservative estimate of the cross-stock momentum unrelated to factor momentum. This is because excluding the symmetric component can eliminate symmetric cross-stock linkages unrelated to factor momentum.¹²

Table 3 shows that the asymmetric component *ASY* alone generates a significant FF7 alpha of 1.02% per month (t -stat=4.75). The results are similar when using the asymmetric correlation in Eq. (6) as the prediction matrix (1.07% per month FF7 alpha, t -stat=5.02). The results corroborate the findings in Section 6.1 that cross-stock momentum is not entirely due to factor momentum.

6.3. Does cross-stock linkage drive factor momentum?

According to Ehsani and Linnainmaa (2022, Eq. (IA5)), the time-series factor momentum profit using F factors is driven by:

$$\frac{1}{F} E \left[\sum_{f=1}^F R_{t-1}^f R_t^f \right], \quad (14)$$

where R_t^f denotes the return of factor f in month t . This corresponds to a trading strategy where the weight on factor f in month t is determined by R_{t-1}^f . We rewrite each factor return using individual stock returns, which implies:

$$\begin{aligned} R_{t-1}^f R_t^f &= \left(\sum_j w_{t-1}^{j,f} R_{t-1}^j \right) \left(\sum_i w_t^{i,f} R_t^i \right) \\ &= \sum_{\substack{j \neq i \\ \text{high } j \rightarrow i \text{ link}}} w_{t-1}^{j,f} w_t^{i,f} R_{t-1}^j R_t^i \\ &\quad + \sum_{\substack{j \neq i \\ \text{low } j \rightarrow i \text{ link}}} w_{t-1}^{j,f} w_t^{i,f} R_{t-1}^j R_t^i \\ &\quad + \sum_{j=i} w_{t-1}^{j,f} w_t^{i,f} R_{t-1}^j R_t^i, \end{aligned} \quad (15)$$

where $w_t^{i,f}$ is the weight of stock i in factor f and R_t^i is stock i 's return in month t .

For each stock pair i and j , there are two lead-lag linkages between the two stocks: the linkage from R_{t-1}^j to R_t^i and the linkage from R_{t-1}^i to R_t^j . We decompose the factor momentum according to whether stock j leads i , according to Definition 1, adjusted by the signs of the two stocks'

¹² For example, shocks to a specific geographic area can potentially generate symmetric lead-lag relation for two stocks in that area, but such shocks may not be factor shocks common to the entire cross-section of stocks.

weights in factor f . Specifically, a high (low) $j \rightarrow i$ link in Eq. (15) is determined by:

$$j \rightarrow i \text{ link} = \begin{cases} \text{high} & \text{if } \text{Cor}(R_{t-1}^j, R_t^i) \text{sgn}(w_{t-1}^{j,f} w_t^{i,f}) \geq \text{Cor}(R_{t-1}^i, R_t^j) \text{sgn}(w_{t-1}^{i,f} w_t^{j,f}) \\ \text{low} & \text{if } \text{Cor}(R_{t-1}^j, R_t^i) \text{sgn}(w_{t-1}^{j,f} w_t^{i,f}) \leq \text{Cor}(R_{t-1}^i, R_t^j) \text{sgn}(w_{t-1}^{i,f} w_t^{j,f}), \end{cases} \quad (16)$$

where $\text{sgn}(x)$ is the sign function that equals 1, 0, or -1 if x is positive, zero, or negative, respectively.¹³ A high $j \rightarrow i$ link is equivalent to a low $i \rightarrow j$ link.

If all the sign functions equal one (i.e., the two stocks' weights in factor f are either both positive or both negative), a high (low) $j \rightarrow i$ link is equivalent to stock j leading (lagging) stock i as defined in Definition 1. In this case, if j leads i , we expect $R_{t-1}^j R_t^i$ to contribute more to factor momentum than $R_{t-1}^i R_t^j$. Therefore, we expect the first term to exceed the second term in Eq. (15) if cross-stock momentum affects factor momentum.¹⁴ When a sign function is negative, the weights of the two stocks have the opposite sign. In this case, a higher cross-stock linkage implies a lower contribution to factor momentum. Therefore, the $j \rightarrow i$ link incorporates the sign functions so the first term is expected to be higher than the second term in Eq. (15) based on cross-stock linkages.

Next, we derive a decomposition analogous to Eq. (15) for the cross-sectional factor momentum. According to Ehsani and Linnainmaa (2022, Eq. (IA3)), the cross-sectional factor momentum is driven by $\frac{1}{F} E[\sum_{f=1}^F (R_{t-1}^f - \bar{R}_{t-1}) R_t^f]$, where $\bar{R}_{t-1} = \frac{1}{F} \sum_{f=1}^F R_{t-1}^f$ is the average factor return at $t-1$. This corresponds to a trading strategy where the weight on factor f in month t is positive if the factor's past return is above average and negative if it is below average. We focus on the following equivalent version of the cross-sectional factor momentum:

$$\frac{1}{F} E \left[\sum_{f=1}^F (R_{t-1}^f - \bar{R}_{t-1}) (R_t^f - \bar{R}_t) \right], \quad (17)$$

which maintains the symmetry between $t-1$ and t to facilitate analyses.¹⁵ The average factor return $\bar{R}_{t-1}$ can be re-written using individual stock returns as $\bar{R}_{t-1} = \sum_i \tilde{w}_{t-1}^i R_{t-1}^i$, where $\tilde{w}_{t-1}^i \equiv \frac{1}{F} \sum_{f=1}^F w_{t-1}^{i,f}$ is the average weight of stock i in the factors.¹⁶ Therefore,

$$(R_{t-1}^f - \bar{R}_{t-1}) (R_t^f - \bar{R}_t)$$

¹³ If the inequality in Eq. (16) holds as an equality, we include the $j \rightarrow i$ link in both the first and second terms in Eq. (15), which does not affect their comparison.

¹⁴ In Eq. (15), the factor weights also affect the first two terms. However, if the factor weights do not change drastically between adjacent months (i.e., $w_{t-1}^{j,f} \approx w_t^{j,f}$ and $w_{t-1}^{i,f} \approx w_t^{i,f}$), the weights in the first two terms are similar. We confirm in Section A.1 of the Online Appendix that variations in factor weights at the monthly frequency only have a minor effect on our empirical results.

¹⁵ Equation (17) is equivalent to $\frac{1}{F} E[\sum_{f=1}^F (R_{t-1}^f - \bar{R}_{t-1}) R_t^f]$ because $\frac{1}{F} E[\sum_{f=1}^F (R_{t-1}^f - \bar{R}_{t-1}) \bar{R}_t] = E[\bar{R}_t (\frac{1}{F} \sum_{f=1}^F R_{t-1}^f - \bar{R}_{t-1})] = E[\bar{R}_t (\bar{R}_{t-1} - \bar{R}_{t-1})] = E[\bar{R}_t \cdot 0] = 0$.

¹⁶ To see this, $\bar{R}_{t-1} = \frac{1}{F} \sum_{f=1}^F R_{t-1}^f = \frac{1}{F} \sum_{f=1}^F \sum_i w_{t-1}^{i,f} R_{t-1}^i = \sum_i R_{t-1}^i (\frac{1}{F} \sum_{f=1}^F w_{t-1}^{i,f}) = \sum_i R_{t-1}^i \tilde{w}_{t-1}^i$.

$$\begin{aligned} &= \left(\sum_j w_{t-1}^{j,f} R_{t-1}^j - \sum_j \tilde{w}_{t-1}^j R_{t-1}^j \right) \left(\sum_i w_t^{i,f} R_t^i - \sum_i \tilde{w}_t^i R_t^i \right) \\ &= \left(\sum_j \tilde{w}_{t-1}^{j,f} R_{t-1}^j \right) \left(\sum_i \tilde{w}_t^{i,f} R_t^i \right) \quad \text{where } \tilde{w}_t^{i,f} \equiv w_t^{i,f} - \tilde{w}_t^i \\ &= \sum_{\substack{j \neq i \\ \text{high } j \rightarrow i \text{ link}}} \tilde{w}_{t-1}^{j,f} \tilde{w}_t^{i,f} R_{t-1}^j R_t^i + \sum_{\substack{j \neq i \\ \text{low } j \rightarrow i \text{ link}}} \tilde{w}_{t-1}^{j,f} \tilde{w}_t^{i,f} R_{t-1}^j R_t^i \\ &\quad + \sum_{j=i} \tilde{w}_{t-1}^{j,f} \tilde{w}_t^{j,f} R_{t-1}^j R_t^j, \end{aligned} \quad (18)$$

where the last step is derived similarly to Eq. (15) but uses $\tilde{w}$ as the weight in the factors. $\tilde{w}$ also replaces w in Eq. (16) to determine whether the linkage from stock j to i is expected to make a high or low contribution to the cross-sectional factor momentum.

Table 5 shows the results for the factor momentum decomposition. It includes the decomposition for the time-series factor momentum in Eq. (15) and the decomposition for the cross-sectional factor momentum in Eq. (18). When using the 15 factors in Ehsani and Linnainmaa (2022), the factor momentum component corresponding to high cross-stock links (first summation in Eqs. (15) and (18)) is significantly higher than the factor momentum component corresponding to low cross-stock links (second summation in Eqs. (15) and (18)). The component from high cross-stock links exceeds that from low cross-stock links by 1.78 (t -stat=3.43) for the time-series factor momentum and by 1.90 (t -stat=3.63) for the cross-sectional factor momentum. Bootstrap confidence intervals also confirm that the factor momentum component from high cross-stock links is significantly higher than the factor momentum component from low cross-stock links.

Under the null hypothesis that the cross-stock linkages do not matter for factor momentum or that the cross-stock linkages are entirely derived from factor momentum, we expect the first and second summations in Eqs. (15) and (18) to be the same. This is because factor momentum likely implies symmetry, as shown in Eq. (13). The null hypothesis is rejected based on Table 5. Therefore, cross-stock momentum is not entirely driven by factor momentum and helps explain factor momentum.

Further, the positive factor momentum is mostly from the component with high cross-stock links. For the time-series factor momentum, the component with high cross-stock links is 1.10 (t -stat=3.78), while the component with low cross-stock links is -0.68 (t -stat=-2.53). The component corresponding to the own stock momentum (last summation in Eq. (15)) is 0.01 (t -stat=0.63). Therefore, own stock momentum does not drive factor momentum, consistent with the finding in Ehsani and Linnainmaa (2022). The result is similar for cross-sectional factor momentum, where the component with high cross-stock links is 1.15 (t -stat=3.88) while the component with low cross-stock links is -0.75 (t -stat=-2.81). Table 5 also shows the factor momentum decomposition using the 43 factors in Arnott et al., 2023 and the results are similar to those using 15 factors.

Table 5**Decompose factor momentum by cross-stock link.**

We decompose the factor momentum into three components according to Eq. (15) for the time-series factor momentum and according to Eq. (18) for the cross-sectional factor momentum. The factor momentum component with a high (low) cross-stock link corresponds to the first (second) term in Eq. (15) and (18). The factor momentum component from the own-stock lead-lag corresponds to the last term in Eq. (15) and (18). $Cor(R_{t-1}^i, R_t^i)$ in Eq. (16), which determines whether the cross-stock link is high or low, is calculated using a five-year lookback window. We use the 15 factors in Ehsani and Linnainmaa (2022) or the 43 factors in Arnott et al. (2023) to construct factor momentum. Following Lo and MacKinlay (1990, Table 3), the results are multiplied by 10,000. t -statistics are in the parentheses. This table also shows a bootstrap 95% confidence interval for "High – Low" (denoted by "Bootstrap CI"). We generate 50,000 bootstrap samples, where each sample is randomly drawn from the "High – Low" data with replacement.

Factor Mom		Own-stock	Cross-stock link		High - Low	Bootstrap CI
		lead-lag	Low	High	High - Low	
15 factors	TimeSeries	0.01 (0.63)	−0.68 (−2.53)	1.10 (3.78)	1.78 (3.43)	[0.82, 2.83]
	CrossSection	0.01 (0.59)	−0.75 (−2.81)	1.15 (3.88)	1.90 (3.63)	[0.93, 2.96]
43 factors	TimeSeries	0.05 (0.84)	−0.75 (−2.52)	1.07 (3.50)	1.83 (3.11)	[0.73, 3.03]
	CrossSection	0.05 (0.84)	−0.92 (−2.89)	1.23 (3.70)	2.15 (3.38)	[0.98, 3.48]

Table 6**Decompose factor momentum by cross-stock link, detailed results by factor.**

This table shows the detailed results factor by factor for the decomposition of time-series momentum using 15 factors in Table 5. Specifically, it shows the three components in Eq. (15) for each factor.

Factor	Own-stock	Cross-stock link		High - Low	Bootstrap CI
	lead-lag	Low	High	High - Low	
Accrual	0.01 (0.37)	−0.63 (−2.06)	0.64 (2.17)	1.27 (2.14)	[0.12, 2.46]
BaB	0.02 (0.95)	−0.41 (−0.78)	1.78 (3.01)	2.19 (3.66)	[1.07, 3.40]
BM	0.02 (1.14)	−0.63 (−1.72)	1.07 (3.23)	1.70 (2.82)	[0.55, 2.91]
CashtoPrice	0.01 (0.68)	−0.58 (−1.68)	1.03 (3.36)	1.61 (2.87)	[0.53, 2.73]
EarningstoPrice	0.01 (0.56)	−0.44 (−1.38)	0.98 (3.58)	1.42 (2.62)	[0.38, 2.50]
Investment	−0.00 (−0.07)	−0.71 (−2.35)	0.62 (2.08)	1.33 (2.25)	[0.19, 2.50]
Ivol	0.02 (1.35)	−0.93 (−2.16)	1.36 (2.25)	2.28 (2.56)	[0.56, 4.03]
Liquidity	−0.00 (−0.12)	−0.61 (−2.21)	0.74 (2.95)	1.35 (2.65)	[0.37, 2.38]
LongtermReversal	0.02 (1.06)	−0.68 (−2.33)	1.12 (3.80)	1.81 (3.42)	[0.82, 2.89]
Momentum	0.03 (1.52)	−0.77 (−1.85)	1.51 (2.51)	2.27 (3.32)	[0.98, 3.67]
NetShareIssuance	0.02 (1.05)	−0.38 (−1.12)	1.13 (3.67)	1.50 (2.50)	[0.35, 2.72]
Profitability	0.00 (0.16)	−0.58 (−1.93)	1.18 (4.26)	1.76 (3.20)	[0.70, 2.86]
QualityminusJunk	0.01 (0.36)	−0.71 (−2.32)	0.92 (3.23)	1.63 (2.82)	[0.52, 2.79]
ShorttermReversal	−0.02 (−1.65)	−1.31 (−2.47)	1.32 (2.10)	2.62 (2.40)	[0.52, 4.79]
Size	0.01 (0.48)	−0.89 (−3.18)	1.13 (4.14)	2.01 (3.95)	[1.05, 3.04]
15 factors combined	0.01 (0.63)	−0.68 (−2.53)	1.10 (3.78)	1.78 (3.43)	[0.82, 2.83]

In addition, Table 6 shows the time-series factor momentum decomposition for each of the 15 factors in Ehsani and Linnainmaa (2022). In other words, we use one factor at a time in Eq. (14) by setting $F = 1$. Table 6 shows that the results in Table 5 hold for each factor individually, where each factor's time-series momentum is statisti-

cally and economically higher for the component with high cross-stock links than the component with low cross-stock links.¹⁷

¹⁷ We also replicate Table 6 using the 43 factors in Arnott et al., 2023 or for the cross-sectional factor momentum. The results are similar and sup-

Table 7**Asymmetric and symmetric linkages between stocks in the same industry.**

This table shows the cross-stock momentum returns based on the asymmetric or symmetric parts of the industry momentum prediction matrix (ASY and SYM in Eq. (1), respectively). We compute the equal-weighted decile return spreads. For the asymmetric component, we additionally show the cross-stock momentum returns where the decile sort uses only stocks whose market capitalizations are in the top 10% (or bottom 90%) within its industry, which are denoted “top size decile stocks” (or “other stocks”). The sample period is 1963–2018 for comparison with factor momentum.

	Excess return	CAPM alpha	FF3 alpha	FF5 alpha	FF7 alpha
Asymmetry					
all stocks	0.47 (5.87)	0.47 (5.83)	0.46 (5.64)	0.44 (5.26)	0.53 (6.32)
top size decile stocks	−0.26 (−1.64)	−0.27 (−1.7)	−0.28 (−1.75)	−0.25 (−1.51)	−0.56 (−3.96)
other stocks	0.66 (6.47)	0.68 (6.57)	0.66 (6.29)	0.63 (5.88)	0.82 (8.25)
Symmetry	0.67 (4.74)	0.73 (5.18)	0.73 (5.11)	0.76 (5.13)	1.07 (9.78)

Our finding that high cross-stock links provide almost all the factor momentum profit is consistent with [Lo and MacKinlay \(1990\)](#), who suggest that positive autocorrelation in index returns “may be attributed solely to the positive cross-autocorrelations across securities.” Our finding is also consistent with [Falck et al., 2022](#), who show that factor momentum is especially anomalous at the one-month lead-lag time scale, because cross-stock momentum is typically the strongest at the one-month horizon.

6.4. Does factor momentum subsume industry momentum?

[Arnott et al., 2023](#) argue that factor momentum subsumes industry momentum. In this subsection, we distinguish between industry momentum and factor momentum by using the asymmetry of the industry linkages.

Let Π_I denote the prediction matrix of industry momentum. For two stocks from different industries, their corresponding entry in Π_I is zero. Following [Eq. \(1\)](#), the asymmetric component ASY and the symmetric component SYM satisfy:

$$\Pi_I = \text{SYM} + \text{ASY}$$

$$\Pi_I^T = \text{SYM} - \text{ASY},$$

where Π_I^T indicates the matrix transposition of Π_I . Therefore, the asymmetric component of industry linkages is $\text{ASY} = (\Pi_I - \Pi_I^T)/2$, and the symmetric component is $\text{SYM} = (\Pi_I + \Pi_I^T)/2$.

[Table 7](#) shows the portfolio return alphas based on ASY of the industry linkages. ASY excludes all symmetric cross-stock linkages, which can include those unrelated to factor momentum. Therefore, the alpha from ASY is a conservative estimate of industry momentum unrelated to factor momentum. [Table 7](#) shows an FF7 alpha of 0.53% per month (t -stat = 6.32).

The evidence in [Table 7](#) indicates that industry momentum is not entirely subsumed by factor momentum. How does this reconcile with [Arnott et al., 2023](#), who find insignificant industry momentum after controlling for factor momentum? In [Sections 6.4.1–6.4.2](#), we reconcile our findings with those in [Arnott et al., 2023](#).

6.4.1. Amplification of misspecification via value weight

First, the asymmetry ASY is misspecified even if we assume the industry linkages are correctly specified. Since $\Pi_I = \text{SYM} + \text{ASY}$, ASY differs from Π_I if SYM is not zero. The difference from Π_I implies that ASY is misspecified relative to Π_I .¹⁸ The misspecification results from excluding SYM to differentiate with factor momentum using asymmetry. Despite the misspecification, ASY may contain sufficient information to generate a significant cross-stock momentum, as shown in [Table 7](#). However, if the misspecification is accentuated, it can overshadow the valid information. We next show that ASY for industry momentum is likely misspecified for large stocks, which is amplified by the value-weighted industry returns studied in [Arnott et al., 2023](#).

Industry linkages use size to determine the ability of a stock's return to predict the future returns of other stocks in the same industry. The returns of larger stocks can be more representative of an industry's return and better forecast the future returns of stocks in the same industry than smaller stocks. The first plot in [Fig. 3](#) shows the overall industry momentum prediction matrix Π_I . We sort the rows and columns of the matrix by size within an industry to focus on potential misspecification associated with size.¹⁹ The plot illustrates the high linkages associated with the lagged returns of large stocks (highest value for the rightmost column).

¹⁸ The symmetry SYM is also misspecified if ASY is not zero. However, we focus on ASY to distinguish industry momentum from factor momentum.

¹⁹ We also sort size without conditioning on industry and the results in [Section 6.4](#) hold. Because stocks are sorted by size in [Fig. 3](#), different industries are scattered. So it is not visible in [Fig. 3](#) that stocks in different industries have zero linkage.

pressed for brevity. For the cross-sectional factor momentum, we still use the average return across 15 or 43 factors as the cutoff for past factor returns when replicating [Table 6](#). Therefore, the detailed cross-sectional factor momentum result for each factor is interpreted as a decomposition of [Eq. \(17\)](#) into individual factors instead of a cross-sectional strategy using only one factor.

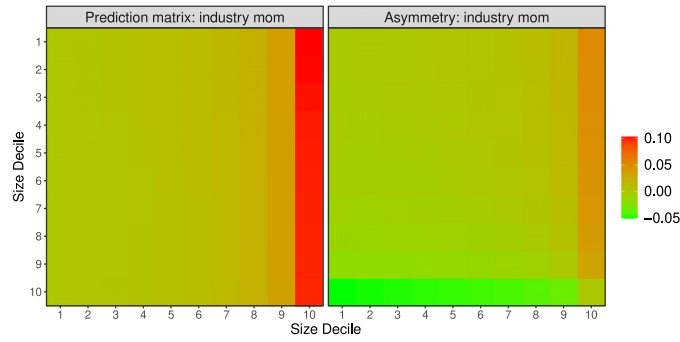


Fig. 3. Industry momentum prediction matrix and its asymmetry. This figure shows the industry momentum prediction matrix and its asymmetric component for stocks sorted by size. Within each industry, we sort stocks by size into deciles (decile 10 contains the largest stocks). The left plot shows the prediction matrix, which is averaged first within the 10×10 size sorts and then over time. The vertical (horizontal) axis corresponds to stocks whose future (past) returns are used in industry momentum. The right plot shows the asymmetric component (ASY in Eq. (1)) for industry momentum. The sample period is 1963–2018 for comparison with factor momentum.

Although large stocks may lead other stocks, one might suspect that their abilities to lead other stocks are not proportional to their sizes. If Π_I specifies excessively positive lead from large stocks to other stocks, its asymmetry prescribes excessively negative lead from other stocks to large stocks due to the subtraction of Π_I^T in ASY. The last plot in Fig. 3 shows ASY for industry linkages. The bottom row of ASY tends to be negative even though Π_I indicates positive industry momentum (all the elements in Π_I are non-negative). The bottom row predicts the future returns of large stocks heavily weighted in value-weighted returns.

To confirm the potential misspecification in the bottom row of ASY, Table 7 shows the industry momentum return from ASY when trading only stocks in the top size decile while still using signals from all stocks. The FF7 alpha is negative, consistent with misspecification for large stocks. When we construct industry momentum from ASY by trading other stocks outside the top size decile, the FF7 alpha is 0.82% per month (t -stat = 8.25), stronger than the FF7 alpha using all stocks.

6.4.2. Reconcile our results and Arnott et al., 2023

We hypothesize that industry momentum is not subsumed by factor momentum if we alleviate the misspecification associated with large stocks. In Panel A of Table 8, we replicate Arnott et al., 2023, Table 4). When controlling factor momentum, the value-weighted industry momentum alpha is insignificant (first row of Panel A), consistent with Arnott et al., 2023. However, if we exclude the top size decile stocks from the industry returns, the value-weighted industry momentum has a significant alpha of 0.26% per month (t -stat = 3.44). When using equal-weighted industry return to alleviate the misspecification of large stocks, the alpha remains significant at 0.34% per month (t -stat = 5.22). The result is similar if we exclude the top size quintile stocks from the value-weighted industry returns or if we exclude the microcap stocks from the equal-weighted industry returns.

In Panel A of Table 8, we follow Arnott et al., 2023 and use the cross-sectional construction of industry momentum that takes long positions in industries with above-median returns in the previous month and short

those with below-median returns. In Panel B, we follow the top-bottom construction of industry momentum in Moskowitz and Grinblatt (1999), which takes long (short) positions in the three industries with the highest (lowest) returns in the previous month. The results are even stronger than those in Panel A. For example, when controlling for factor momentum, the value-weighted industry momentum is insignificant but increases to 0.61% per month (t -stat = 3.89) when stocks in the top size decile are excluded from the value-weighted industry returns.

The results show that factor momentum does not subsume industry momentum. Our paper reveals two reasons why industry momentum appears insignificant in Arnott et al., 2023 when controlling for factor momentum. First, the symmetric component of industry linkages can be tricky to distinguish from factor momentum because factor momentum also prescribes symmetric linkages. Second, the asymmetric component of industry linkages is misspecified, which is more pronounced for large stocks. Analyzing value-weighted industry return amplifies the misspecification, which can overwhelm the valid information in the asymmetry.

7. Time-varying linkages across stocks

The cross-stock linkages captured by the PP prediction matrix vary over time faster than the linkages in previous studies. Fig. 4 shows that the linkage from the PP prediction matrix has lower autocorrelation than all the existing linkages. For lags beyond five years, the autocorrelation of the PP linkage is virtually zero. By contrast, the other cross-stock linkages tend to be highly stable. The most stable linkage, geographic location, has a five-year autocorrelation of 95%, followed by the industry linkage, which has a five-year autocorrelation of 88%. The technology link has the lowest five-year autocorrelation of 43% among the existing linkages. Even though the cross-stock linkages specified in previous studies are persistent over time, Section A.4 of the Online Appendix confirms that their return predictability largely comes from stock pairs with high PP linkages,

Table 8
Industry momentum, weighting the biggest stocks less.

We regress industry momentum on factor momentum. Following [Arnott et al. \(2023\)](#), Table 4), we control for FF5 factors and use the cross-sectional construction of industry momentum that takes long positions in industries with above-median returns in the previous month and short those with below-median returns in Panel A. In Panel B, we use the top-bottom construction of industry momentum following [Moskowitz and Grinblatt \(1999\)](#) and [Arnott et al. \(2023\)](#), which takes long (short) positions in the three industries with the highest (lowest) returns in the previous month. We use the value-weighted returns of all stocks (or stocks outside the top size decile or quintile) in an industry as the value-weighted (VW) industry return. Size deciles and quintiles are defined within an industry. We use the equal-weighted returns of all stocks (or non-microcap stocks) in an industry as the equal-weighted (EW) industry return. This table reports the monthly alpha of industry momentum ($\hat{\alpha}$) and the slope coefficient of factor momentum ($\hat{b}_{fmom}$). We use the 20 industries in [Moskowitz and Grinblatt \(1999\)](#). Due to the availability of factor data, the sample period is 1963–2018.

Panel A. Cross-sectional construction of industry momentum				
Dependent variable	$\hat{\alpha}$	$\hat{b}_{fmom}$	FF5	R^2
VW Industry Momentum	0.12 (1.57)	0.40 (16.62)	Y	31.2%
VW Industry Momentum (exclude top size decile)	0.26 (3.44)	0.55 (22.49)	Y	46.2%
VW Industry Momentum (exclude top size quintile)	0.27 (3.50)	0.55 (22.59)	Y	47.1%
EW Industry Momentum	0.34 (5.22)	0.55 (26.68)	Y	54.0%
EW Industry Momentum (exclude microcap)	0.31 (4.06)	0.51 (20.76)	Y	41.1%
Panel B. Top-bottom construction of industry momentum				
Dependent variable	$\hat{\alpha}$	$\hat{b}_{fmom}$	FF5	R^2
VW Industry Momentum	0.21 (1.42)	0.82 (17.12)	Y	32.0%
VW Industry Momentum (exclude top size decile)	0.61 (3.89)	1.05 (20.81)	Y	43.1%
VW Industry Momentum (exclude top size quintile)	0.56 (3.45)	1.04 (19.96)	Y	41.1%
EW Industry Momentum	0.64 (4.78)	1.14 (26.61)	Y	53.9%
EW Industry Momentum (exclude microcap)	0.49 (3.17)	1.10 (21.89)	Y	44.7%

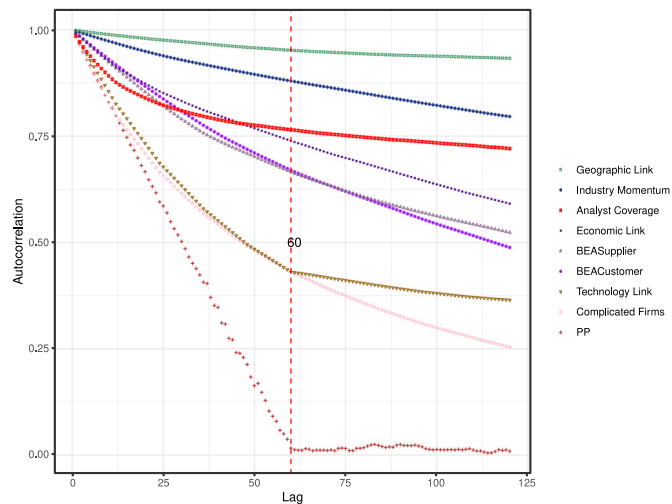


Fig. 4. Autocorrelation of prediction matrices. This figure shows the autocorrelation of the PP prediction matrix and the prediction matrices of eight existing cross-stock momentums up to 120-month lags. The matrices are vectorized for computing the autocorrelation.

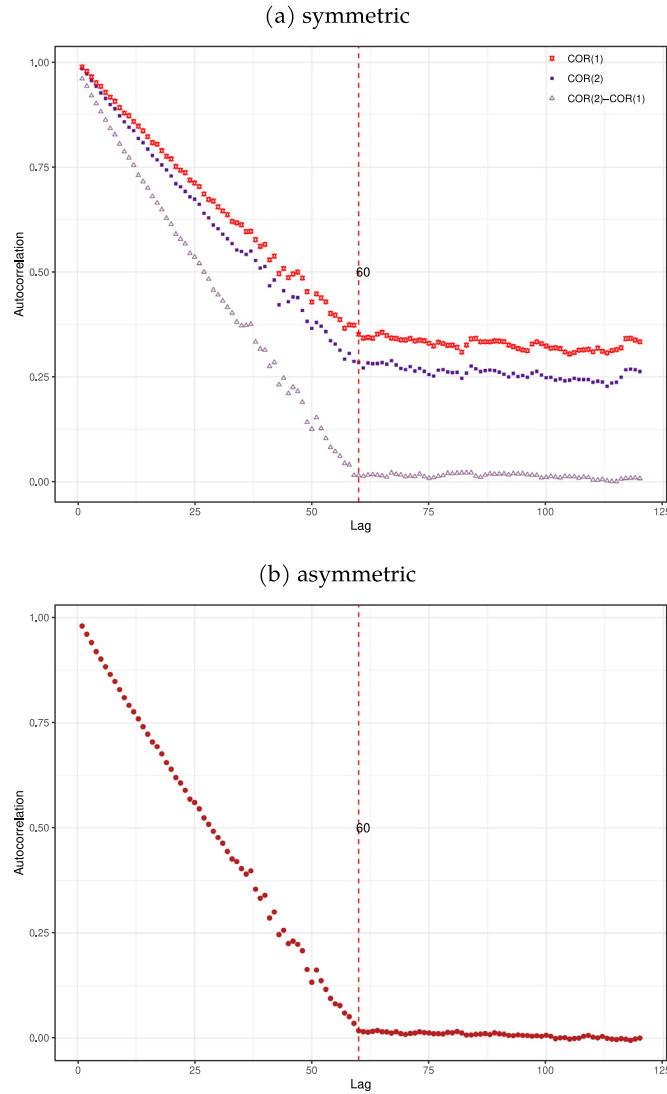


Fig. 5. Autocorrelation of the components of cross-stock linkages. This figure shows the autocorrelation of the components of cross-stock linkage up to 120-month lags. Fig. 5a shows the autocorrelation for the symmetric components: $COR(1)$, $COR(2)$, and $COR(2) - COR(1)$ in Eq. (4). Fig. 5b shows the autocorrelation for the asymmetric component denoted “asymmetric cor” in Eq. (6). The matrix for each component is vectorized for computing the autocorrelation. The sample period is 1931–2018.

which inherit the low persistence of the PP prediction matrix.

To examine the return predictability from the time-varying component, we project $\hat{\Pi}_t$ on the prediction matrix five years ago $\hat{\Pi}_{t-5y}$:

$$\hat{\Pi}_t = a + b \times \hat{\Pi}_{t-5y} + \varepsilon_t. \quad (19)$$

Following Eq. (6) of Keloharju et al. (2021), we use the residual ε_t as the transitory component.²⁰ Table 9 shows

²⁰ We vectorize the prediction matrix $\hat{\Pi}$ when running the regression in Eq. (19). We use the Fama and MacBeth (1973) cross-sectional regression to accommodate the large number of observations—the number of observations for the regression in each month is on the order of $O(n^2)$ when there are n stocks. Otherwise, a pooled regression using data from all months has $O(n^2T)$ observations where T is the number of months. When averaging the monthly Fama-MacBeth estimates, we use only the past months to avoid potential look-ahead biases.

the PP return using the transitory component ε_t as the prediction matrix. This table also shows the PP returns using $\hat{\Pi}_t$ as the prediction matrix for comparison. The returns from the transitory component are almost identical to the returns using $\hat{\Pi}$. The estimated slope coefficient b in Eq. (19) is only 0.02 on average, implying that the lagged prediction matrix does not affect $\hat{\Pi}$ much. Therefore, the results from the residual and $\hat{\Pi}$ are similar, which is consistent with the low autocorrelation of the PP prediction matrix in Fig. 4.

7.1. Why are the lead-lag linkages time-varying?

To understand the low persistence of the lead-lag linkage across stocks, Fig. 5a shows the persistence of $COR(1)$ and $COR(2)$ in Eq. (4) using all stocks in the PP prediction matrix. The five-year autocorrelation coefficients of $COR(1)$

Table 9
Transitory component of $\hat{\Pi}$.

This table is similar to Table 2 except that we use the transitory component of the prediction matrix $\hat{\Pi}$ to construct the PP return. The transitory component is from the residual of an auto-regressive regression of $\hat{\Pi}$ on its five-year lag in Eq. (19). The regression uses only past data to avoid the look-ahead bias and is estimated using the Fama and MacBeth (1973) method. This table also shows the PP returns using $\hat{\Pi}$ in the same sample period.

	Excess returns	CAPM alpha	FF3 alpha	FF3 + m alpha	FF3 + mr alpha	FF5 alpha	FF7 alpha
Transitory	0.88 (6.15)	0.89 (6.13)	0.86 (5.92)	0.95 (6.39)	1.40 (9.98)	1.00 (4.77)	1.52 (7.79)
$\hat{\Pi}$	0.88 (6.15)	0.89 (6.13)	0.87 (5.94)	0.95 (6.38)	1.41 (10.07)	1.01 (4.81)	1.52 (7.85)

and $COR(2)$ are around 0.3, higher than near-zero autocorrelation of the PP linkage in Fig. 4. $COR(1)$ and $COR(2)$ capture contemporaneous common shocks affecting different stocks. Therefore, the contemporaneous linkages across stocks have some persistence.

However, what matters for the lead-lag linkage is not just the contemporaneous linkage. According to Eq. (4), the symmetric component of lead-lag linkages depends on $COR(2) - COR(1)$, the difference between longer- and shorter-horizon co-movements. Fig. 5a shows that $COR(2) - COR(1)$ has almost zero autocorrelation at the five-year lag. This indicates the cancellation of the persistent components in $COR(2)$ and $COR(1)$. It might be easier for investors to learn about the linkages that do not vary much over time. Therefore, common shocks associated with such persistent linkages likely are incorporated into the stock prices simultaneously upon arrival and do not substantially drive the lead-lag linkages.²¹

Fig. 5b shows the persistence of the asymmetric component of lead-lag linkage. Its five-year autocorrelation is almost zero, similar to the symmetric component. According to Eq. (6), the asymmetric component depends on the difference between $Cor(r_t^i, r_{t-1}^j)$ and $Cor(r_t^j, r_{t-1}^i)$, which are already lead-lag correlations and have near-zero five-year autocorrelation themselves.²²

The prediction matrices in previous studies of cross-stock momentum are often designed to capture common shocks across stocks. A high persistence of the contemporaneous linkage can be consistent with a low persistence of the lead-lag linkage if investors recognize the persistent component of the contemporaneous linkage. Our results indicate that time series properties of co-movements matter for cross-stock momentum.

8. Conclusion

We dissect cross-stock momentum by first characterizing the lead-lag linkages across stocks theoretically. The

cross-stock linkages consist of an asymmetric component and a symmetric component. The asymmetric component can be decomposed into a high lead-lag linkage from a leader stock to a laggard stock and a low lead-lag linkage from the laggard stock to the leader stock, which reflect cross-sectional variations in lead-lag linkages. The symmetric component depends on long-run and short-run contemporaneous co-movements, capturing their time-series nature.

We use the principal portfolio (PP) method in Kelly et al., 2023 to construct data-driven linkages and analyze the counterparts of our theoretical characterization. The data-driven linkages generate significant return predictability. Singular value decomposition (SVD) allows a parsimonious representation of the cross-firm relation using no more than $O(n)$ complexity. The leading terms of SVD empirically confirm that the cross-sectional asymmetry and the time-series properties of co-movements are important for cross-stock momentum. Consistent with the dimension reduction, PPs based on the leading terms of SVD capture substantial cross-stock predictability.

We use the cross-sectional asymmetry in lead-lag linkages and find that cross-stock momentum is not purely due to factor momentum. We show Arnott et al., 2023's finding that industry momentum stems from factor momentum is due to the misspecification of the asymmetry in industry linkages for large stocks exacerbated by the value-weighted industry returns they analyze. After mitigating the misspecification, industry momentum becomes significant when we replicate their analysis.

Additionally, we analyze the sources of factor momentum profits using the methodology of Ehsani and Linnainmaa (2022) and Arnott et al., 2023 but disaggregate to the stock level. We find that the factor momentum profit is mostly from the high cross-stock links. The low cross-stock links contribute negatively to factor momentum. The evidence is consistent with the hypothesis in Lo and MacKinlay (1990) that momentum at the portfolio level "may be attributed solely to the positive cross-autocorrelations across securities." We exploit the asymmetry so that the cross-autocorrelations driving factor momentum are unlikely to be derived from factor momentum.

Our analysis shows that the cross-stock linkages mainly responsible for return predictability vary over time at approximately the business cycle frequency. The persistent long-run co-movements are already incorporated into the short-run co-movements. The cancellation of persistent components in the long-run and short-run co-movements

²¹ As an additional illustration, Figure A1 of the Online Appendix shows the persistence of $COR(2)$ and $COR(1)$ for stocks in the same industry. The five-year autocorrelation coefficients of $COR(1)$ and $COR(2)$ are around 0.6, which resembles the high persistence of the industry momentum prediction matrix in Fig. 4. However, $COR(2) - COR(1)$ still exhibits near-zero persistence.

²² In Fig. 5b, we do not separately plot the persistence of $Cor(r_t^i, r_{t-1}^j)$ and $Cor(r_t^j, r_{t-1}^i)$ because their autocorrelation coefficients for horizons at five years and more are virtually indistinguishable from those in Fig. 5b.

explains our finding that the data-driven linkages are less persistent than those in previous studies.

Our results suggest a more prominent role for asymmetry in cross-stock linkages. The linkages in previous studies of cross-stock momentum tend to either be symmetric or assume that asymmetry is determined by equal or size weight, which we find can sometimes contain misspecification. Refining the asymmetry when constructing linkages can enhance cross-stock momentum and distinguish it from factor momentum. Our analysis can be applied to firm networks to understand their implications for asset pricing and beyond.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jfineco.2023.103716](https://doi.org/10.1016/j.jfineco.2023.103716).

References

- Ali, U., Hirshleifer, D., 2020. Shared analyst coverage: Unifying momentum spillover effects. *J. Financ. Econ.* 136 (3), 649–675.
- Arnott, R. D., Kalesnik, V., Linnainmaa, J. T., 2023. Factor momentum. *Rev. Financ. Stud.* 36 (8), 3034–3070.
- Burt, A., Hrdlicka, C., 2021. Where does the predictability from sorting on returns of economically linked firms come from? *J. Financ. Quant. Anal.* 56 (8), 2634–2658.
- Campbell, J.Y., Lo, A., MacKinlay, A.C., 1997. *The Econometrics of Financial Markets*. Princeton University Press.
- Carhart, M.M., 1997. On persistence in mutual fund performance. *J. Finance* 52 (1), 57–82.
- Chinco, A., Clark-Joseph, A.D., Ye, M., 2019. Sparse signals in the cross-section of returns. *J. Finance* 74 (1), 449–492.
- Cohen, L., Frazzini, A., 2008. Economic links and predictable returns. *J. Finance* 63 (4), 1977–2011.
- Cohen, L., Lou, D., 2012. Complicated firms. *J. Financ. Econ.* 104 (2), 383–400.
- Ehsani, S., Linnainmaa, J.T., 2022. Factor momentum and the momentum factor. *J. Finance* 77 (3), 1877–1919.
- Falck, A., Rej, A., Thesmar, D., 2022. Is factor momentum greater than stock momentum? *J. Invest. Strateg.* 10 (4), 43–68.
- Fama, E., French, K., 1993. Common risk factors in the returns on stocks and bonds. *J. Financ. Econ.* 33, 3–56.
- Fama, E., MacBeth, J., 1973. Risk, return and equilibrium: empirical tests. *J. Polit. Economy* 81, 607–636.
- Fama, E.F., French, K.R., 2008. Dissecting anomalies. *J. Finance* 63 (4), 1653–1678.
- Fama, E.F., French, K.R., 2015. A five-factor asset pricing model. *J. Financ. Econ.* 116 (1), 1–22.
- Feng, G., Giglio, S., Xiu, D., 2020. Taming the factor zoo: a test of new factors. *J. Finance* 75 (3), 1327–1370.
- Freyberger, J., Neuhierl, A., Weber, M., 2020. Dissecting characteristics nonparametrically. *Rev. Financ. Stud.* 33 (5), 2326–2377.
- Gu, S., Kelly, B., Xiu, D., 2020. Empirical asset pricing via machine learning. *Rev. Financ. Stud.* 33 (5), 2223–2273.
- Hou, K., 2007. Industry information diffusion and the lead-lag effect in stock returns. *Rev. Financ. Stud.* 20 (4), 1113–1138.
- Hou, K., Xue, C., Zhang, L., 2020. Replicating anomalies. *Rev. Financ. Stud.* 33 (5), 2019–2133.
- Jackson, M.O., 2008. *Social and Economic Networks*. Princeton University Press.
- Jin, Z., Li, F. W., 2021. Geographic links and predictable returns. *SSRN* 3617417.
- Kelly, B., Malamud, S., Pedersen, L. H., 2023. Principal portfolios. *J. Finance* 78 (1), 347–387.
- Kelly, B. T., Moskowitz, T. J., Pruitt, S., 2021. Understanding momentum and reversal. *J. Financ. Econ.* 140 (3), 726–743.
- Kelly, B.T., Pruitt, S., Su, Y., 2019. Characteristics are covariances: a unified model of risk and return. *J. Financ. Econ.* 134 (3), 501–524.
- Keloharju, M., Linnainmaa, J.T., Nyberg, P., 2021. Long-term discount rates do not vary across firms. *J. Financ. Econ.* 141 (3), 946–967.
- Kim, S., Korajczyk, R.A., Neuhierl, A., 2020. Arbitrage portfolios. *Rev. Financ. Stud.* 34 (6), 2813–2856.
- Kogan, L., Papanikolaou, D., Seru, A., Stoffman, N., 2017. Technological innovation, resource allocation, and growth. *Q. J. Econ.* 132 (2), 665–712.
- Kozak, S., Nagel, S., Santosh, S., 2020. Shrinking the cross-section. *J. Financ. Econ.* 135 (2), 271–292.
- Lee, C.M.C., Shi, T., Sun, S.T., Zhang, R., 2022. Production Complementarity and Momentum Spillover Across Industries. Working Paper.
- Lee, C.M.C., Sun, S.T., Wang, R., Zhang, R., 2019. Technological links and predictable returns. *J. Financ. Econ.* 132 (3), 76–96.
- Lettau, M., Pelger, M., 2020. Factors that fit the time series and cross-section of stock returns. *Rev. Financ. Stud.* 33 (5), 2274–2325.
- Lewellen, J., 2002. Momentum and autocorrelation in stock returns. *Rev. Financ. Stud.* 15 (2), 533–564.
- Lo, A.W., MacKinlay, A.C., 1990. When are contrarian profits due to stock market overreaction? *Rev. Financ. Stud.* 3 (2), 175–205.
- Menzly, L., Ozbas, O., 2010. Market segmentation and cross-predictability of returns. *J. Finance* 65 (4), 1555–1580.
- Moskowitz, T.J., Grinblatt, M., 1999. Do industries explain momentum? *J. Finance* 54 (4), 1249–1290.
- Parsons, C.A., Sabbatucci, R., Titman, S., 2020. Geographic lead-lag effects. *Rev. Financ. Stud.* 33 (10), 4721–4770.
- Rapach, D.E., Strauss, J.K., Zhou, G., 2013. International stock return predictability: what is the role of the united states? *J. Finance* 68 (4), 1633–1662.
- Rossi, A., 2018. Predicting Stock Market Returns with Machine Learning. Working paper.
- Scherbina, A., Schlusche, B., 2020. Follow the leader: using the stock market to uncover information flows between firms. *Rev. Finance* 24 (1), 189–225.
- Shumway, T., 1997. The delisting bias in CRSP data. *J. Finance* 52 (1), 327–340.